Learned control of urinary reflexes in cattle to help reduce greenhouse gas emissions

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Abstract

Indiscriminate voiding of excreta by cattle contributes to greenhouse gas (GHG) emissions and soil and water contamination[1,2]. Emissions are higher in animal-friendly husbandry offering cattle more space[2] - a trade-off we call the 'climate killer conundrum'. Voiding in a specific location (latrine) would help resolve this dilemma by allowing ready capture and treatment of excreta under more spacious farming conditions. For urination, toileting requires self-control and coordination of a complex chain of behaviors including awareness of bladder fullness, overriding of excretory reflexes, selection of a latrine and intentional relaxation of the external urethral sphincter[3]. Attempts to train toileting in cattle have so far been only partly successful[4-6], even though their excretion and associated neurophysiological control are similar to those in species capable of toileting[3]. Similarly, very young infants have been considered incapable of self-initiated voiding, but they can be taught with extensive training[7]. Using a backward chaining, reward-based training procedure, we here show that cattle can control their micturition reflex and use a latrine for urination. Such self-control provides evidence that animals can learn to respond to and reveal internal experiences via appropriately trained operant behaviors, thereby providing another way to explore their subjective states.

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